

Day - 17

RANGE() & LOOPS

in Python



 **SKILL COURSE**



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What is range()?

Layman Explanation

- range() is like telling Python:
 “Start from here...
 Go till there...
 Move step by step.”
- It helps you generate a sequence of numbers that you can loop through.

• Technical Definition

- range() generates a sequence of integers.
 Used mostly with **for loops**.
- Syntax : range(start, stop, step)
- **Where:**
 - **start** → where to begin (default 0)
 - **stop** → where to stop (not included)
 - **step** → increment (default 1)



Why Range + Loop is Important for Data Analysts?

- Repeat tasks specific number of times
- Process rows one-by-one
- Automate calculations
- Create dashboards / reports
- Generate dummy data
- Create IDs automatically
- Perform time-based looping
- Clean multiple columns with indexes



Basic Example



Print 1 to 5

```
for i in range(1, 6):  
    print(i)
```



Print 0 to 4

```
for i in range(5):  
    print(i)
```

More Examples

Even Numbers

```
for i in range(0, 20, 2):  
    print(i)
```

Odd Numbers

```
for i in range(1, 20, 2):  
    print(i)
```

Countdown Using Range

```
for i in range(10, 0, -1):  
    print(i)
```

Looping Through Items Using Index

```
items = ["Pen", "Book", "Laptop"]
```

```
for i in range(len(items)):  
    print(i, items[i])
```



Sum of List Using Range

```
nums = [10, 20, 30, 40]
```

```
total = 0
```

```
for i in range(len(nums)):
```

```
    total += nums[i]
```

```
print("Total:", total)
```



Clean Multiple Values Using Index

```
cities = [" mUMbai", " pune ", " DELhi"]  
  
for i in range(len(cities)):  
    cities[i] = cities[i].strip().title()  
  
print(cities)
```



Nested Loops with Range

```
for i in range(1, 4):  
    for j in range(1, 4):  
        print(i, j)
```

